

Problem Set 3

Spring 2026

Due: 11:59 pm Monday, 13 Apr

To facilitate grading and timely feedback
please note that all submissions are through Gradescope.

Solve each problem on a new page.

Please clearly identify collaborator names and the resources used for each problem.

1. For each n , let $p(n)$ be the fraction of strings in $\{0,1\}^n$ that represent a zero-input Turing machine that halts when run starting from a blank tape. This means that a $1 - p(n)$ fraction of these strings are either invalid Turing machine representations, or represent machines that run forever.
 - (a) Describe an algorithm $M(i, n)$ to compute asymptotically good lower bounds $p_i(n)$ that converge to $p(n)$ from below as $i \rightarrow \infty$. This means that $p_i(n) \leq p(n)$, and that for every n there are only finitely many i for which $p_i(n) \neq p(n)$.
 - (b) Why doesn't the algorithm in part (a) immediately allow you to compute $p(n)$?
 - (c) Show that the function $p(n)$ is not computable.
 - (d) Show that there is no algorithm $N(i, n)$ that is like $M(i, n)$ from part (a), but computes upper bounds rather than a lower bounds.
2. For each of the following languages, determine whether the language is decidable and whether it is recognizable, and prove your answer.
 - (a) $L_3 = \{(\langle M_1 \rangle, \langle M_2 \rangle) \mid M_1() \text{ or } M_2() \text{ accepts (or both)}\}$
 - (b) $L_4 = \{\langle M \rangle \mid L(M) \text{ is infinite and } \overline{L(M)} \text{ is also infinite}\}$

3. Let $\langle M \rangle$ be the encoding of a Turing Machine. We define the following languages:

$$HALT = \{\langle M \rangle \mid M() \text{ halts}\}$$

$$FiniteCell = \{\langle M \rangle \mid \text{the total number of tape cells } M() \text{ ever visits is finite}\}.$$

- (a) Is $FiniteCell \subseteq HALT$? (If yes, explain why; if no, describe an element of $FiniteCell$ not in $HALT$.)
 - (b) Is $HALT \subseteq FiniteCell$? (If yes, explain why; if no, describe an element of $HALT$ not in $FiniteCell$.)
 - (c) Is $FiniteCell \leq_T HALT$? (If yes, describe a reduction; if no, explain why not.)
 - (d) Is $HALT \leq_T FiniteCell$? (If yes, describe a reduction; if no, explain why not.)
4. Let $\langle M \rangle$ be the encoding of a Turing Machine. We define the following languages:

$$ChillyHALT = \{(\langle M \rangle, x) \mid M^{HALT}(x) \text{ halts}\}$$

$$FinLeft = \{\langle M \rangle \mid \text{the max distance tape head moves to the left of initial location is finite}\}$$

$$FinHALT = \{\langle M \rangle \mid M(x) \text{ halts for at most finitely many inputs } x\}.$$

For this question you can use the fact that $ChillyHALT$ is Turing-equivalent to $FinHALT$.

- (a) Show that $FinLeft \leq_T ChillyHALT$.
 - (b) Show that $ChillyHALT \leq_T FinLeft$.
5. Given a string $x \in \{0,1\}^n$, recall that $K(x)$ is the length of the shortest program P (in some universal programming language) such that $P() = x$. We proved in class that the function K is uncomputable. Note that in our current definition the program P takes no input.
- (a) Show that $K \leq_T HALT$.
 - (b) Let $K^{HALT}(x)$, the “Chilly Kolmogorov complexity” of x , be the length of the shortest program P such that $P^{HALT}() = x$. By adapting the proof given in class for the uncomputability of K , show that $K^{HALT} \not\leq_T HALT$.
 - (c) *Bonus.* Show that for every n , there exists a string $x \in \{0,1\}^n$ such that $K^{HALT}(x) = O(\log n)$ even though $K(x) = \Omega(n)$.